Pandas DataFrame

Units 6–10 · syllabus 2024 · module marks 15, doubled to 30

HOW THIS PAPER WORKS

The syllabus gives this module 15 marks. This paper gives **twice** that (30 marks). The questions use the physics data named in the syllabus: capacitor charging, phone-sensor readings, and pendulum and spring experiments.

No past paper of this exact course was found, so the questions use similar CUFYUGP papers. **PYQ** = adapted from a similar paper, named with its code, term and question number. **Instructor** = written from the syllabus, with no close match.

Similar papers used: CSC1MN106 Computer Fundamentals with MS Excel, SPSS (Nov 2024); STA2MN101 Probability Theory I (Apr 2025); STA3MN201 Statistical Inference using R (Nov 2025); MAT4CJ205 Python and SageMath (Apr 2026).

Time	2 hours
Maximum	30 marks

Section A: answer all ten. Section B: answer any two of three. Show the code you would run. Where a table is given, write the output as a table.

Section A · Short answer

Answer all. Each question carries 2 marks. Total 20.

1. Write the code that creates a DataFrame `df` with two columns, `Length` and `Period`, from the lists `[0.25, 0.50, 1.00]` and `[1.00, 1.42, 2.01]`.

PYQ A26 Q16 · adapted · U6

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2. For the DataFrame of question 1, state what `df.shape`, `df.columns` and `df.dtypes` return.

Instructor · U7

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3. Long-format data has columns `trial`, `capacitor` and `voltage`. Explain what `df.pivot(index='trial', columns='capacitor', values='voltage')` produces.

PYQ CSC1MN106 N24 A7 · adapted · U7

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4. What is the difference between `df.sort_values('Period')` and `df.sort_index()`? Which one sorts by row labels?

Instructor · U7

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5. A sensor file has the missing value `-999` in some rows. Write one line that replaces every `-999` with `NaN`, and one line that drops every row that still contains a `NaN`.

Instructor · U8

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6. Explain what `df.describe()` reports for a numeric column, and state what `df['Period'].idxmax()` returns.

PYQ CSC1MN106 N24 B18 · adapted · U8

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7. Write the code for the mode and the count of non-missing values in a column named `mass`.

PYQ STA3MN201 N25 B13 · adapted · U9

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8. A student computes the covariance and the correlation of mass and extension for a spring experiment. State the range of the correlation coefficient and say which of the two quantities has units.

PYQ STA2MN101 A25 B15 · adapted · U9

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9. Explain what `df['Period'].quantile(0.5)` returns, and write one line that applies `np.sqrt` to every value in a column `Period` using `apply()`.

Instructor · U9

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10. A table has a column `material` with values `steel`, `copper`, `rubber` and a column `k`. Write the code that finds the mean value of `k` for each material using `groupby()`.

Instructor · U10

Section B · Descriptive

Answer any two of the three. Each carries 5 marks. Total 10.

11. Capacitor charging. Three capacitors are charged and the voltage is logged every second. The data is read from `charge.csv` with columns `time`, `C1`, `C2` and `C3`, and some cells are empty. Write the code that (a) reads the file, (b) fills the empty cells with the column mean, and (c) prints the correlation between `time` and `C1`. Explain why the correlation alone does not show that the voltage follows an exponential law.

PYQ CSC1MN106 N24 B12 · adapted · U8, U17

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12. Two tables are given. `students` has columns `name` and `group`; `marks` has columns `name` and `score`. Explain the difference between `pd.merge(students, marks, on='name')` and `pd.concat([students, marks], axis=1)`. Give a short example where the two give different results.

PYQ CSC1MN106 N24 A8 · adapted · U10

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13. A phone sensor records acceleration once every 0.1 second for one minute. Write code that (a) builds a time series with `pd.date_range` using the correct frequency, (b) makes it a Series indexed by time, and (c) selects the readings between the 10th and the 20th second. Explain the role of the index in step (c).

Instructor · U10

COVERAGE

UNIT	TOPIC	QUESTIONS
6	Create a DataFrame	1
7	Attributes, pivot, sort, sort by labels	2, 3, 4
8	Missing data, combining, describe, min and max index	5, 6, 11
9	count, mode, covariance, correlation, quantiles, pipe, apply	7, 8, 9, 11
10	Aggregation, groupby, merging, concatenating, time series	10, 12, 13

Total marks: 20 (Section A) + 10 (Section B) = 30. Every topic of units 6–10 is the main point of at least one question. **`pipe()`** is asked in Section A, question 9.